

Skills Summary

Rational and Irrational Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Proving a number is rational

Definition: a number is *rational* when it can be written as one integer over another. **The proof is the fraction** — find it and the argument is finished.

Three families that are always rational: integers (put them over one), terminating decimals (put them over a power of ten), and repeating decimals (use the subtract-the-copy trick).

Example: Show that 0.24 is rational.

Step 1. The decimal stops, so it is already a fraction in disguise — read the last digit's place value and write it down.

Step 2. The last digit sits in the hundredths place, so $0.24 = \frac{24}{100}$, and dividing top and bottom by four gives lowest terms.

$$\Rightarrow 0.24 = \frac{6}{25}$$

Try it: Show that 0.45 is rational.

check: $\frac{9}{20}$

Skill 2 — Deciding whether a root is rational

Rule: the radical sign decides nothing. $\sqrt{N}$ is rational exactly when N is a perfect square (or a fraction of two perfect squares); otherwise it is irrational.

π is irrational, and so is every non-terminating, non-repeating decimal.

Example: Is $\sqrt{225}$ rational?

Step 1. Rational-or-not for a root is a perfect-square question, so I test whether the radicand is a square rather than reaching for a calculator.

Step 2. Counting up the squares reaches the radicand exactly, so the root is a whole number and a whole number is itself over one.

$$\Rightarrow \sqrt{225} = 15, \text{ rational}$$

Try it: Is $\sqrt{196}$ rational? If so, give its value.

check: 14

(!) Watch out: a root sign does not make a number irrational, and a decimal that runs on does not either. $\sqrt{225}$ is rational; $0.\bar{3}$ is rational. Only a decimal that never ends *and* never repeats is irrational.

Skill 3 — Placing an irrational number on a number line

Method: trap the irrational root between the two perfect squares on either side of the radicand. Their roots are the consecutive whole numbers it lies between; then squeeze with tenths and hundredths to place the point.

Example: Place $\sqrt{20}$ on a number line.

Step 1. No fraction exists for this root, so the useful answer is an interval and a rounded decimal — trap it between neighbouring perfect squares.

Step 2. The radicand sits between the squares of the fourth and fifth whole numbers, so the root is between them; squeezing with hundredths lands just under one half of the way up.

$$\Rightarrow \sqrt{20} \approx 4.47$$

Try it: Place $\sqrt{30}$ to the nearest hundredth.

check: 5.48